

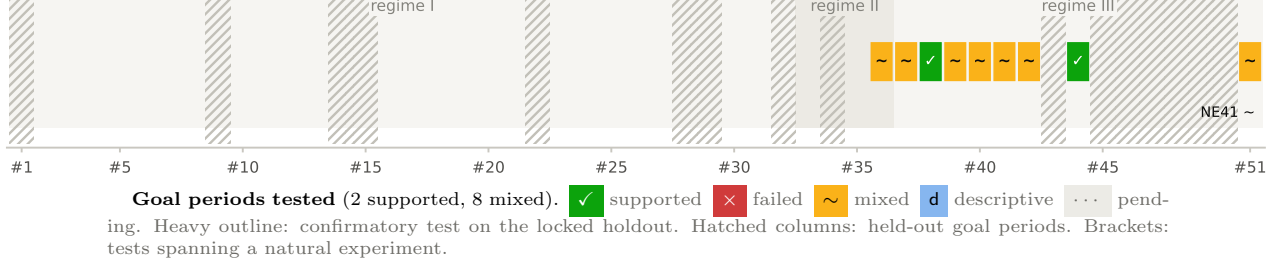
H44: Erasure makes agents busy but unproductive

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 42% Faithfulness 2/5 Usefulness 3/5

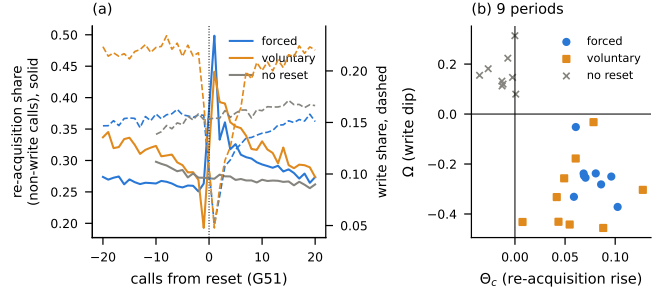
Bottom line. A context wipe triggers an orderly re-reading burst and a ~26% write dip that cost 4–11% of output and break most command loops, but it does not heat the agent up or make it more susceptible to the room.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

When the scaffold wipes an agent’s context window at the 41-record cap (memory kept), does the agent thrash: re-read instead of producing, act more randomly, and listen harder to new room content? I aligned 21,165 forced and 16,357 voluntary resets with no-reset pseudo-erasures over 1.24M model calls in nine non-holdout regime-III periods. A synthetic validation came first. Re-acquisition is real. The read share doubles at the first post-reset call (G51: 0.22 to 0.47). Among non-write calls it stays up over five calls ($\Theta_c = +0.079$, 9/9 periods). Writes fall 26%, work commits 38%, real failures rise 1 point. There is no temperature pulse and no extra pull from new room items (RR 0.97), but coupling to pre-reset items drops (ratio 0.60). Erasures break command loops (OR 0.11). Holdout not run.



Observables. (a) G51: share of non-write calls that re-acquire (solid) and write share (dashed) vs calls from a forced reset, a voluntary reset, or a no-reset boundary. Re-reading doubles at the first call and decays with a ~7-call tail. Writes drop 68% at the first call and then ramp back. (b) All nine periods: forced and voluntary resets sit at $\Theta_c > 0$, $\Omega < 0$. Pseudo-erasures sit at $\Theta_c \leq 0$ and $\Omega > 0$, because writes ramp up through every segment.

PHYSICAL MODEL

Model 02 as a kinetic multinomial spin. Each model call has a category s_t (write, run, read, look, GUI, room read, talk, idle, ...). Its update has a baseline field h^0 , a room drive h^{room} , a context-held task field $h^{\text{ctx}}(n)$ and a persistence $J(n)$, where n counts calls since the last reset; an erasure sets $n \rightarrow 0$. R_{nw} is the read share of non-write calls, c' an agent $\times$ previous-category cell, W the write share. Model 04 asks which store restores output.

$$P(s_{t+1}=c) \propto e^{h_c^0 + h_c^{\text{room}} + h_c^{\text{ctx}}(n) + J(n)\mathbb{I}[c=s_t]}$$

$$h^{\text{ctx}}(n) = h_{\infty}^{\text{ctx}} (1 - e^{-n/\ell})$$

$$\Theta_c = \sum_{c'} w_{c'} [R_{\text{nw}}(\text{post} | c') - R_{\text{nw}}(\text{far} | c')]$$

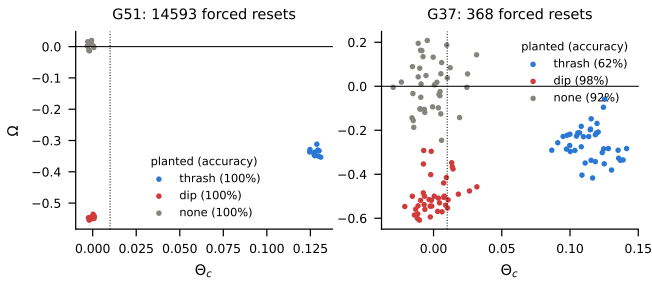
$$\Omega = W_{\text{post}}/W_{\text{far}} - 1, \quad \Delta V = \sum_{k \leq 20} (W_k - W_{\text{far}})$$

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Useful, and more concrete than the physics. The 41-record cap costs each period 4–9% of its write calls and 5–11% of its work commits. After a wipe the agent re-reads, fails a few more commands, and rebuilds its task from its files over 3–7 calls. Memory notes written at the wipe do not help, and it does not turn to the room. Three things an operator can use: the cap length is a cost/benefit dial (it also breaks most command loops); the cheapest recovery hint points the agent at its working files (re-reading them resumes writing 2.6 calls sooner than looking at the screen); and a freshly wiped agent is no easier for other agents to steer. “Temperature pulse” did not survive, and “thrash” overstates it: the re-reading is orderly.

RESULTS VS PREDICTIONS

Prediction	Observed (pooled, 9 periods)	
P1 re-acquisition $\Theta > 0$	$\Theta_c + 0.079$ [0.071, 0.086], 9/9; pseudo -0.011	✓
P2 entropy, switching up	$\Delta H -0.027$; $\Delta\sigma +0.018$; ℓ tail 5–7 calls	×
P3 write dip -25 to -60%	$\Omega -26\%$ (8/9); commits -38%	✓
P4a coupling to pre-reset items down	ratio 0.60 [0.44, 0.81]	✓
P4b susceptibility to new items up	RR 0.97 [0.91, 1.04]; G51 0.90	×
P4c content pull up	D -0.026 (n.s.); G51 -0.027	×
P5 forced $\geq \frac{1}{2}$ voluntary	0.079 vs 0.057; pre-trend = no-reset band	✓
G51 artifact-first recovers fastest	-2.6 calls to first write vs screen; dose $\rho -0.03$	✓
G38 resets break loops	recur 16% vs 72%, OR 0.08	✓
G36 onset larger, learning trend	onset 0.075 vs 0.080; no trend	×



Second figure. Synthetic validation on real call skeletons: planted thrash, pure output dip and no effect, classified by Θ_c (dotted: floor 0.01) and Ω . G51 counts: 100% correct. G37 counts: thrash found in 62% (otherwise the dip is missed), never a false thrash. The raw, unconditioned Θ would have called the pure dip “thrash”.

GOAL PERIODS IN DETAIL

- G36 ~ mixed (replication mixed (re-acquisition up, writes down, no entrop...
- G37 ~ mixed (re-acquisition up, write dip not significant, no entropy/swi...
- G38 ✓ supported (replication supported (re-acquisition up, writes down, s...
- G39 ~ mixed (re-acquisition up, writes down, no entropy/switching rise)
- G40 ~ mixed (re-acquisition up, writes down, no entropy/switching rise)
- G41 ~ mixed (re-acquisition up, writes down, no entropy/switching rise)
- G42 ~ mixed (re-acquisition up, writes down, no entropy/switching rise)
- G44 ✓ supported (re-acquisition up, writes down, switching up)
- G51 ~ mixed (replication mixed (re-acquisition up, writes down, switching...
- NE41 ~ mixed (re-acquisition and output dip supported in 9/9 and 8/9 perio...

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Call categories come from in-memory regex classification of commands and from GUI action types, with no hand-labelled validation. GUI browsing cannot be separated from production. Writes ramp up through every 40-call segment, so the dip size depends on the reference window. Θ_c conditions on first-order state only. The re-acquisition paths are chosen by the agents, so the path results are observational. Jev’s 5-min windows cannot resolve the 1–5-call burst; they label post-reset windows as *more* executing, as H39 saw. G37 is underpowered. Nine periods times about 12 statistics, uncorrected. Holdout not run.

ROUND 2

Validate the classifier. Find which files are re-read (the semantic content of the context store). Find the segment length that maximizes output per call. Price loop breaking against re-acquisition. Run `confirm.py` after disclosure.

- **Essence.** how much of an agent’s productive state lives in its context window rather than in memory or artifacts, and what an operator pays, and gains, each time the scaffold wipes it.
- **H44-R1** Validate the call classifier.
- **H44-R2** The sawtooth and the best cap.
- **H44-R3** What exactly is re-read.
- **H44-R4** Loop breaking as a lever.
- **H44-R5** Why no susceptibility rise.
- **H44-R6** Run `confirm.py` after committing the card and disclosing reuse.